Md. Shahariar Hossain

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Education

M.Sc. in Data Science Expected May 2027
BRAC University, Dhaka, Bangladesh | Current CGPA: 4.00/4.00

B.Sc. in Electrical and Electronic Engineering May 2026
BRAC University, Dhaka, Bangladesh | CGPA: 3.60/4.00
Project: A Quadruped Robot for Real-Time Inspection and Responsive Operations.

Publications

Journal Article

[1] **M. S. Hossain**, M. T. Hassan, and A. Islam, "G-buffer-guided feature modulation for lightweight rendered image super-resolution," *Array*, vol. 32, Art. no. 101324, 2026. doi: 10.1016/j.array.2026.101324.

Conference Papers

[1] **M. S. Hossain**, M. T. Hassan, and M. S. Hossain, "Biologically Inspired Network Design to Train Only Through Forward Pass," in *Proc. 2nd Int. Conf. Information and Communication Technology (ICICT)*, Dhaka, Bangladesh, 2024, pp. 130–134. doi: 10.1109/ICICT64387.2024.10839663.

[2] **M. S. Hossain**, M. S. Hossain, and M. B. Karim, "Training StyleGAN2 on BWW-Texture Dataset: Generating Textures Through Transfer Learning," in *Proc. Int. Conf. Information and Communication Technology for Sustainable Development (ICICT4SD)*, Dhaka, Bangladesh, 2023, pp. 341–345. doi: 10.1109/ICICT4SD59951.2023.10303468.

Accepted Conference Paper

[1] **M. S. Hossain**, R. M. Abrar, S. A. Akib, T. Ahmed, S. Alam, and A. M. Oni, "Kinematic Analysis, Trajectory Generation, and Low-Cost Hardware Evaluation for a Quadruped Robot Leg," IEEE RAAICON, accepted, 2026.

Book

[1] M. S. Hossain and **M. S. Hossain**, *Mohabishwer Mohajatra* [The Great Journey of the Universe]. 1st ed., March 2021, 2nd printing, February 2023. 284 pp. ISBN 978-984-95421-7-9. Science book on cosmology and the history of the universe.

Research Experience

Research Assistant Sep. 2026–Present
BSRM School of Engineering, BRAC University
Supervisor: Md Arif Billah, Ph.D., Assistant Professor, EEE

- Derived state-space models and feedback control equations for swarm robotics and evaluated nonlinear dynamics in MATLAB simulations.
- Designed robots in Autodesk Fusion and developed a five-agent swarm platform using low-cost edge hardware.
- Implemented reinforcement learning and latent-state world models for adaptive trajectories and autonomous control.

Directed Research (EEE 498) Jun. 2025–May 2026
BRAC University | Rendered Image Super-Resolution
Supervisor: Amirul Islam, Ph.D., Associate Professor, EEE

- Designed TensorFlow models using convolution, attention, and FiLM to combine low-resolution RGB renders with high-resolution albedo for $2\times$, $3\times$, and $4\times$ upscaling.
- Built a dataset of 11,000 paired RGB and albedo images from Unreal Engine renders and evaluated auxiliary-channel fusion through ablations, retraining, and PSNR/SSIM comparisons.

Final Year Design Project (EEE 499) Jun. 2025–May 2026
BRAC University | Quadruped Robot for Real-Time Inspection
Advisory panel: Academic Technical Committee, EEE

- Built a quadruped platform with onboard perception for remote inspection of confined, hazardous sites.
- Developed reinforcement learning, inverse kinematics, and walking and trotting gaits. Modeled the robot in Autodesk Fusion and simulated it in ROS/Gazebo and NVIDIA Isaac Sim.
- Coordinated a four-member team, presented design reviews to faculty and industry partners, and led safety analysis incorporating ISO 45001 considerations.

Honors and Awards

- **Special Recognition Award**, BRAC University Freshers Orientation, Sep. 2026.
- **Champion, Project Showcase**, EWU RobotFest, Jun. 2026. Team award for the quadruped inspection robot.
- **First Prize**, BCSIR National Science Fair, Feb. 2020. Project lead, "Electricity-Generating Road."
- **Best Creative Writer**, Science Division, Jatiya Shrijanshil Medha Annyashan, 2020.
- **Regional Champion and National Finalist**, Bangladesh Physics Olympiad, Rajshahi, 2016.
- National finalist distinctions: Astrophysics, top 5 (2016–2021), Physics, top 10 (2020), Junior Science, top 20 (2017), and National High School Programming Contest, top 20 (2016).

Leadership, Mentoring, and Outreach

- **Member**, IEEE Student Branch and Robotics Club, BRAC University, 2022–present. Machine learning and vision demonstrations, embedded systems workshops, and robotics outreach.
- **Co-founder**, Science Galaxy STEM Club, Rajshahi University School & College, 2016–2020. Organized science fairs, talks, and student research showcases.
- **Mentor**, Bangladesh Physics, Math, and Astrophysics Olympiad training workshops for high school students, 2018–2020.

Technical Skills

- **Programming and ML:** Python, TensorFlow, PyTorch, NumPy, C/C++.
- **Robotics and simulation:** ROS, Gazebo, NVIDIA Isaac Sim, Webots, MATLAB, PSpice, Proteus.
- **Design and graphics:** Autodesk Fusion, Altium, AutoCAD, Adobe Illustrator, Blender, Unreal Engine.
- **Platforms and tools:** Linux, Arduino, Raspberry Pi, ATmega32, LaTeX, Microsoft Office.